Curriculum Vitae

Final-year MEng student at Southampton, predicted First and graduating in June 2027. Experience in scientific Python, open-source development and robotics research. Seeking a graduate software engineering role.

contact@ramisbhatty.com GitHub ↗️ LinkedIn ↗️

Education

University of Southampton · Sep 2023 – Jun 2027
MEng Computer Science with Artificial Intelligence · Predicted First
Netcraft Prize: Top 10 in Part I and Part II across Computer Science and Software Engineering degree programmes.

Bournemouth School · Sep 2016 – Jun 2023
A-levels: Maths A*, Computer Science A*, Further Maths A, Physics A.
Ten GCSEs at grade 9 and one at grade 8, including Maths, Computer Science and English.

Full education record →

Experience

Summer Research Intern — University of Southampton, Soorati Lab · Jul – Sep 2026
Python, ROS 2, NumPy, SciPy, MuJoCo, Docker · GitHub

- Developed a LiDAR perception and object-tracking pipeline for assistive robotics, extending existing software to run on an onboard Jetson computer.
- Built simulation and recorded-data tools for repeatable evaluation, and documented setup, testing and remaining limitations for the lab handover.

Internship notes →

Projects and open-source contributions

aeon-neuro: EEG Data Input · Sep 2026 – present
Python, MNE, MNE-BIDS, pytest · GitHub

- Developed EEG data-loading and conversion tools for an open-source Python toolkit, focusing on reliable handling of scientific recordings and metadata.
- Contributed merged changes through code review, automated testing and API documentation, collaborating with maintainers to integrate new functionality into an existing codebase.

Cell Segmentation for Microfluidic Migration Analysis · Sep 2025 – May 2026
Python, PyTorch, OpenCV, NumPy, SciPy, PyQt5 · GitHub

- Built a modular Python framework comparing classical computer vision and deep learning for cell segmentation, combining image processing, model training and experimental evaluation.
- Developed a desktop application for inspecting segmentation results and exporting cell measurements, using background processing to keep the interface responsive.

Project write-up →

CQL Programming Language · Mar – May 2025
Haskell, Alex, Happy

- Collaborated on a SQL-inspired language for CSV data, implementing an interpreter and input checks alongside a language manual and VS Code syntax-highlighting extension.

Technical skills

- **Languages:** Python, Java, SQL, Haskell, C#, C, JavaScript/TypeScript
- **Tools and practices:** Git, Linux, Docker, pytest, unit testing, REST APIs, Agile, LaTeX
- **Personal technical work:** Self-hosts services on a Linux server, including a web frontend for locally deployed LLMs on a separate CUDA-enabled GPU machine. Practical experience with networking, REST API integration and system administration, alongside debugging and rebuilding open-source Java applications.